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Bus Seat Reservation Program

Final Project Programming
Fundamentals

Group H

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Group Members:

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Deliverables for Bus Seat Reservation Program:

1. Objective:

To create a simple C++ program that allows users to view, book, and cancel bus seats. The program should save the seat status to a file so that bookings are retained even after closing the program.

2. Problem Statement:

Manual bus seat management can be confusing and error-prone, often leading to double bookings. A simple computer program is needed to efficiently track available seats and maintain persistent records of bookings.

3. Proposed Solution / Methodology:

- Represent the bus seats using a 2D array of size 10×4.
 - Present a menu to the user with options: view seat map, book a seat, cancel a seat, or exit.
 - Booking a seat sets its value to 1 and cancellation sets it to 0.
 - At program start, load seat data from `seats.txt`. If the file does not exist, initialize all seats to empty.
 - On exit, save the current seat data back to `seats.txt`, ensuring persistence for the next session.
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4. List of Concepts Used:

- 2D Arrays
- Loops (for and do-while)
- Conditional Statements (if-else)
- File Handling (ifstream and ofstream)
- Basic Input/Output (cin and cout)

5. Main Program Code (.cpp file):

```
#include <iostream>
#include <fstream>
using namespace std;

int main() {
    const int rows = 10;
    const int cols = 4;
    int seats[rows][cols];

    // Load from file
    ifstream fin("seats.txt");
    if (fin.is_open()) {
        for (int i = 0; i < rows; i++)
            for (int j = 0; j < cols; j++)
                fin >> seats[i][j];
        fin.close();
    } else {
        for (int i = 0; i < rows; i++)
            for (int j = 0; j < cols; j++)
                seats[i][j] = 0;
    }

    int choice;
    do {
        cout << "\n===== BUS SEAT RESERVATION =====\n";
        cout << "1. View Seat Map\n2. Book a Seat\n3. Cancel a Seat\n4. Exit\n";
```

```

cout << "Enter option: ";
cin >> choice;

if (choice == 1) {
    cout << "\n    BUS SEAT MAP\n";
    cout << " (0=Empty, 1=Booked)\n\n";
    cout << "    LEFT  RIGHT\n    1 2  3 4\n\n";
    for (int i = 0; i < rows; i++) {
        cout << "Row " << i + 1 << " ";
        cout << seats[i][0] << " " << seats[i][1] << " ";
        cout << seats[i][2] << " " << seats[i][3] << endl;
    }
} else if (choice == 2) {
    int r, c;
    cout << "\nEnter row (1-10): "; cin >> r;
    cout << "Enter column (1-4): "; cin >> c;
    r--; c--;
    if (r >= 0 && r < rows && c >= 0 && c < cols) {
        if (seats[r][c] == 0) {
            seats[r][c] = 1;
            cout << "Seat booked successfully!\n";
        } else {
            cout << "Seat already booked!\n";
        }
    } else {
        cout << "Invalid seat number!\n";
    }
} else if (choice == 3) {
    int r, c;
    cout << "\nEnter row (1-10): "; cin >> r;
    cout << "Enter column (1-4): "; cin >> c;
    r--; c--;
    if (r >= 0 && r < rows && c >= 0 && c < cols) {
        if (seats[r][c] == 1) {
            seats[r][c] = 0;
            cout << "Seat cancelled successfully!\n";
        } else {
            cout << "Seat is already empty!\n";
        }
    } else {
        cout << "Invalid seat number!\n";
    }
} else if (choice == 4) {
    ofstream fout("seats.txt");
    for (int i = 0; i < rows; i++) {
        for (int j = 0; j < cols; j++)
            fout << seats[i][j] << " ";
    }
}

```

```

        fout << "\n";
    }
    fout.close();
    cout << "\nData saved. Exiting...\n";
} else {
    cout << "Invalid option. Try again.\n";
}

} while (choice != 4);

return 0;
}

```

6. Sample Input & Output (Working Example):

Input:

```

1    (View seat map)
2    (Book a seat)
Row: 3
Column: 2
1    (View seat map again)
3    (Cancel a seat)
Row: 3
Column: 2
1    (View seat map again)
4    (Exit)

```

Output:

```

BUS SEAT MAP
(0=Empty, 1=Booked)

LEFT  RIGHT
1 2  3 4

Row 1 0 0  0 0
Row 2 0 0  0 0
Row 3 0 0  0 0
...
Seat booked successfully!

```

```
BUS SEAT MAP
Row 3 0 1 0 0
...
Seat cancelled successfully!
BUS SEAT MAP
Row 3 0 0 0 0
Data saved. Exiting...
```

7. Conclusion (What was learned):

Through this project, we learned how to use 2D arrays to represent structured data, implement a menu-driven program, and handle file input/output for data persistence. It also taught basic problem-solving techniques in C++ for real-world applications like seat reservation systems.